

# Coloring random subgraphs

|         |                |                      |              |
|---------|----------------|----------------------|--------------|
| OPG-827 | L1 - Tractable | Partiellement résolu | Graph Theory |
|---------|----------------|----------------------|--------------|

## Énoncé original (anglais, reproduction mot pour mot)

If  $G$  is a graph and  $p \in [0,1]$ , we let  $G_p$  denote a subgraph of  $G$  where each edge of  $G$  appears in  $G_p$  with independently with probability  $p$ .

Problem Does there exist a constant  $c$  so that  $\mathbb{E}(\chi(G_{1/2})) > c \frac{\chi(G)}{\log \chi(G)}$ ?

Source et attribution. Énoncé reproduit depuis UnsolvedMath, jeu de données officiel sous licence CC BY 4.0.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Jeu de données: <https://huggingface.co/datasets/ulamai/UnsolvedMath>